LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

DEPARTMENT OF ECONOMICS

Complexity Adjusted Comparisons of LPs and VARs: Evidence from Monte-Carlo Simulations

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1. Introduction

To infer statements about the dynamic response to exogenous shocks, the estimation of the corresponding impulse response has relied on two fundamental approaches, either through vector autoregressive (VAR) or Local Projection (LP) models. Since the introduction of the latter, an extensive literature has emerged discussing the relative merits of the two. A first strand of literature has established that, under relatively general conditions, LPs and VARs estimate the same population impulse responses (Fusari et al., 2026; Plagborg-Møller & Wolf, 2021). However, in the discussion of finite-sample behavior, LPs were often viewed as less sensitive to misspecification (Jordà, 2005; Ramey, 2016), a property that Olea et al. (2024) subsequently formalise, whereas VARs were seen as potentially more statistically efficient (Li et al., 2024; Olea et al., 2025). Ludwig (2026) offers an analytical explanation of the observed finite-sample bias-variance tradeoff. He shows that LPs can be represented as a sequence of VAR estimations increasing in lag length as the estimated horizon increases, and thus argues for a renewed thinking about how we can compare LPs and VARs, not by lag length, but by effective complexity.

Our paper asks two questions. First, under correct specification, where does a fixed LP estimator sit relative to the full complexity-adjusted VAR frontier, and does the conventional equal-lag ranking survive once that frontier is made visible? Second, once controlled functional misspecification is introduced through a nonlinear DGP, does any genuine LP advantage emerge, and if so, does it manifest in point estimation, in inference, or both?

The first part of this paper operationalizes Ludwig (2026)'s central implication through a complexity-adjusted benchmarking exercise in which a fixed LP(4) estimator is compared to progressively higher-order VAR specifications that proxy increasing effective dynamic flexibility (De Graeve & Westermark, 2025). This comparison is conducted across empirically relevant sample sizes (see e.g. Herbst & Johannsen, 2024) to assess how the resulting bias-variance tradeoff depends on the available data.

The second part introduces controlled functional misspecification. While the baseline experiment studies complexity-adjusted LP-VAR comparisons under correctly specified linear dynamics, the main practical argument for LPs concerns their robustness when the estimated linear model is only an approximation to the true DGP (Montiel Olea & Plagborg-Møller, 2021). Crucially, Plagborg-Møller & Wolf (2021) establish that both LP and VAR estimate the same best linear approximation to the true impulse response in population, regardless of whether the DGP is linear. To study whether this population equivalence translates into symmetric finite-sample performance, the paper augments the baseline VAR(4) with a mean-zero quadratic term in lagged state variables, and examines how the relative performance of LP(4), equal-lag VAR(4), and higher-order VARs changes as the degree of nonlinearity increases.

For each estimator, horizon, DGP, and sample size, we compute bias, variance, mean squared error, and empirical coverage rates. Bias and variance describe the finite-sample tradeoff in point estimation, while coverage assesses whether the corresponding inference procedures remain reliable under increasing functional misspecification.

Our findings suggest that under correct specification, the conventional point-estimation advantage of equal-lag Vector Autoregressions (VARs) persists, but this superiority is largely an artifact of mismatched model complexities. By locating a fixed $LP(4)$ estimator on the complexity-adjusted VAR frontier, we reveal that it operates inherently at the level of a highly parameterized system, intersecting the VAR sweep at orders higher than $q \approx 15$. However, evaluating performance solely through point estimation obscures a critical divergence in inference: local projections (LPs) maintain robust, near-nominal coverage across all persistence regimes, whereas the parametric efficiency of VARs comes at the cost of severe inferential fragility, leading to dramatic under-coverage near the unit root boundary.

When controlled functional misspecification is introduced via a nonlinear data-generating process featuring a quadratic lag term, a genuine LP advantage emerges exclusively in terms of inferential stability rather than point estimation. Because both estimators target the same best linear approximation in population, their finite-sample divergence in point estimation is driven purely by variance. Consequently, the equal-lag $VAR(4)$ continues to strictly dominate the $LP(4)$ reference line in finite-sample mean squared error. However, the nonlinear term induces conditional heteroskedasticity in the projection residuals. The true merit of the direct projection method under functional misspecification thus lies in its robust inference. $LP(4)$ preserves near-nominal confidence interval coverage using heteroskedasticity-robust standard errors, while the coverage of homoskedastic VAR delta-method intervals substantially deteriorates and undercovers.

The remainder of the paper is structured as follows. Section 2 provides a theoretical overview of vector autoregressions and local projections. It also reviews the empirical literature regarding their bias-variance trade-off at equal lag lengths and discusses their asymptotic and finite-sample equivalence. Section 3 presents our baseline simulation study under correct specification. Section 4 extends this framework by introducing dynamic misspecification. Section 5 concludes.

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2. Theoretical Background

Impulse response functions can be defined nonparametrically as the difference between two conditional forecasts of the outcome variable y_{t+h} (Koop et al., 1996). Both forecasts condition on the same vector of controls $\mathbf{x}_t$ and differ only in the time-t realization of the variable of interest s_t: one sets $s_t = s_0 + \delta$, perturbing the baseline value s_0 by a shock of size δ, while the other leaves it at s_0. The conditioning vector $\mathbf{x}_t$ collects the information used to identify the response, the lags of the system and depending on the identification scheme, contemporaneous controls. Following Ramey (2016), a shock here denotes an exogenous, unanticipated movement that is uncorrelated with the controls in $\mathbf{x}_t$ and with all other structural shocks. The difference in the projected paths at horizon h then identifies the causal effect of the shock in population,

$$\mathcal{IR}(h,\delta) \equiv \mathbb{E}[\, y_{t+h} \mid s_t = s_0 + \delta;\, \mathbf{x}_t] - \mathbb{E}[\, y_{t+h} \mid s_t = s_0;\, \mathbf{x}_t] \,. \tag{1}$$

Vector Autoregressions and Local Projections serve as two complementary approaches to estimating impulse responses in finite samples. The former, introduced by Sims (1980), parametrizes the joint dynamics of all variables in a single autoregressive system and impulse responses are then obtained as iterated forecasts from the estimated companion matrix. The latter, established by Jordà (2005), takes a direct forecasting approach by estimating the responses at each horizon independently via OLS, projecting the future outcome directly onto current and lagged values of the data. The two approaches compound estimation error differently across horizons and, at equal lag order, occupy opposite ends of a bias-variance spectrum in finite samples (Li et al., 2024; Olea et al., 2025). Local projections tend toward higher variance, vector autoregressions toward higher bias under misspecification. More recently, however, Ludwig (2026) shows that the equal-order comparison is itself misleading. At a given lag order p, LP(p) is strictly more general than VAR(p), so the conventional bias-variance trade-off conflates the choice of estimator with the choice of model complexity.

2.1 Introduction to VARS (1.5 - 2 pages)

Following Lütkepohl (2005), we start by defining a zero mean VAR(p) population model of the form where the outcome of interest, y_t, is a $K \times 1$ random vector, A_i are fixed $K \times K$ coefficient matrices and u_t are a K dimensional white noise processes with $\mathbb{E}\left[u_t\right] = 0$, $\mathbb{E}\left[u_t u_t{}'\right] = \Sigma_u$, i.e. being a nonsingular covariance matrix, and $\mathbb{E}\left[u_t u_s{}'\right] = 0$ for $t \neq s$. Instead of writing out the full lag structure model the model, we can rewrite the full VAR(p) in its corresponding VAR(1) companion form such that

$$Y_t = \mathbf{A}Y_{t-1} + U_t \tag{2}$$

where

$$Y_t := \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}_{Kp\times 1}, \mathbf{A} := \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}_{Kp\times Kp}, U_t := \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{Kp\times 1}.$$

Assuming Y_t satisfies the stability condition such that all roots, z, of the characteristic equation lie strictly outside the complex unit circle, the process is covariance stationary and admits a Wold moving-average representation:

$$det(I_{Kp} - \mathbf{A}z) \neq 0 \text{ for } |z| \leq 1 \iff Y_t = \sum_{l=0}^{\infty} \mathbf{A}^l U_{t-l}. \tag{3}$$

Restricting attention to the top K rows of the Wold MA representation yields the moving-average representation for y_t:

$$y_t = \sum_{\ell=0}^{\infty} \Psi_\ell \, u_{t-\ell}, \qquad \Psi_\ell := [\mathbf{A}^\ell]_{1:K,\,1:K} \tag{4}$$

where Ψ_h is the reduced-form impulse response function at horizon h, measuring the response of y_t to a reduced-form innovation u_{t-h}. It corresponds to the upper-left $K \times K$ block of $\mathbf{A}^h$, which follows directly from the fact that $U_{t-i} = (u_{t-i}, 0, \ldots, 0)'$ has nonzero entries only in its first K positions. Since the components of u_t are generally contemporaneously correlated, $\Sigma_u = \mathbb{E}(u_t u_t') \neq I_K$, a unit shock to one component of u_t will in general be accompanied by simultaneous movements in the other components. Consequently, the reduced-form impulse responses Ψ_h do not admit a structural interpretation. Structural identification is achieved by imposing restrictions on the impact matrix B in $u_t = B\varepsilon_t$, where $\varepsilon_t \sim (0, I_K)$ are mutually uncorrelated structural shocks, so that $\Sigma_u = BB'$.

The most common approach is recursive Cholesky identification (Ramey, 2016), which restricts B to be lower triangular with positive diagonal entries, uniquely recovering B as the Cholesky factor of Σ_u. The recursive ordering imposes a causal structure: the variable ordered first responds contemporaneously to all shocks, while the variable ordered last is insulated from all contemporaneous feedback. The diagonal entries b_{ii}, $i = 1, \ldots, K$, measure the direct impact of structural shock i on variable i, while the off-diagonal entries b_{ij}, $1 \leq j < i \leq K$, capture the contemporaneous spillover from structural shock j to variable i. Substituting $u_t = B\varepsilon_t$ into (4) yields the structural moving-average representation

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \, \varepsilon_{t-\ell}, \qquad \Theta_\ell := \Psi_\ell B, \qquad B = \begin{pmatrix} b_{11} & 0 & \cdots & 0 \\ b_{21} & b_{22} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ b_{K1} & b_{K2} & \cdots & b_{KK} \end{pmatrix} \tag{5}$$

where the (i, j)-th element of Θ_h measures the response of variable i to a unit innovation in structural shock j at horizon h. The structural impulse response of variable i to shock j is thus

$$\theta_h^{ij} = [\Psi_h B]_{ij} \tag{6}$$

where $[\,\cdot\,]_{ij}$ denotes the (i, j)-th element of a matrix.

From the structural moving-average representation, the structural impulse response is the dynamic causal effect of a one-time, unit-variance shock:

$$\theta_h^{ij} = \frac{\partial \, \mathbb{E}[y_{i,t+h} \mid \varepsilon_{j,t}]}{\partial \, \varepsilon_{j,t}} = [\Theta_h]_{ij} = [\Psi_h B]_{ij} \tag{7}$$

which is exactly the population object $\mathrm{IR}(h, \delta)$ of Equation 1, specialised to a structural innovation of unit size in shock j.

Estimation of the structural impulse response $\hat{\theta}_h^{ij}$ proceeds in two stages. In the first stage, the stacked VAR coefficient matrix $A := [A_1, \ldots, A_p]$ of dimension $K \times Kp$ is estimated equation-by-equation via OLS. Assuming that the underlying process satisfies the stability condition and that the innovations u_t form a strictly stationary martingale difference sequence with a positive definite covariance matrix and finite fourth moments, this estimator is asymptotically normal (Hansen, 2022). Specifically, it yields

$$\sqrt{T} \left(\mathrm{vec}(\hat{A}) - \mathrm{vec}(A) \right) \xrightarrow{d} \mathcal{N}(0, \; V_\alpha) \,, \tag{8}$$

where the general asymptotic covariance matrix takes the robust sandwich form $V_\alpha = \overline{\Gamma}^{-1} \Omega \overline{\Gamma}^{-1}$.

Here, $\overline{\Gamma} = I_K \otimes \Gamma$, with $\Gamma = \mathbb{E}(Z_t Z_t')$ being the second-moment matrix of the stacked regressors $Z_t = (y_{t-1}', \ldots, y_{t-p}')'$, and $\Omega = \mathbb{E}(u_t u_t' \otimes Z_t Z_t')$. If the innovations are conditionally homoskedastic, Ω simplifies to $\Sigma_u \otimes \Gamma$, which reduces the asymptotic variance to the standard form $\Gamma^{-1} \otimes \Sigma_u$ (Lütkepohl, 2005). Under these standard conditions, the equation-by-equation OLS estimator is asymptotically efficient. However, the generalized covariance structure V_α remains crucial to ensure valid inference when the data-generating process exhibits conditional heteroskedasticity.

In the second stage, structural objects are recovered from the reduced-form estimates. The impact matrix $\hat{B}$ is the Cholesky factor of $\hat{\Sigma}_u$, and reduced-form impulse responses are obtained via the recursion

$$\hat{\Psi}_h = \sum_{\ell=1}^{\min(h,p)} \hat{A}_\ell \, \hat{\Psi}_{h-\ell}, \qquad \hat{\Psi}_0 = I_K. \tag{9}$$

so that the structural impulse response estimator $\hat{\theta}_h^{ij} = g(\hat{\alpha}, \hat{\sigma})$, with $\alpha = \mathrm{vec}(A)$ and $\sigma = \mathrm{vech}(\Sigma_u)$, is a smooth nonlinear function of the first-stage estimates,

$$\hat{\theta}_h^{ij} = \left[\hat{\Psi}_h \, \hat{B}\right]_{ij}. \tag{10}$$

Since g is differentiable in $(\hat{\alpha}, \hat{\sigma})$, the delta method yields

$$\sqrt{T}\left(\hat{\theta}_h^{ij} - \theta_h^{ij}\right) \xrightarrow{d} \mathcal{N}\left(0,\ \nabla g(\alpha,\sigma)' \, \Sigma_{\alpha,\sigma} \, \nabla g(\alpha,\sigma)\right), \tag{11}$$

where $\Sigma_{\alpha,\sigma}$ is the joint asymptotic covariance matrix of $(\hat{\alpha}', \hat{\sigma}')'$ and the Jacobian $\nabla g(\alpha, \sigma)$ admits a closed-form expression (Lütkepohl, 1990). The resulting confidence intervals are valid pointwise in h under stationarity, but not uniformly over the persistence of the process nor over growing horizons, motivating a direct examination of finite-sample behaviour.

The favourable asymptotic properties established above are qualified by three interrelated problems that arise in the sample sizes typical of macroeconomic applications. that arise in the sample sizes typical of macroeconomic applications. First, estimation error propagates through iteration. Because $\hat{\Psi}_h$ is the upper-left block of $\hat{\mathbf{A}}^h$, error in $\hat{A}$ enters the impulse response through powers of the companion matrix and is compounded multiplicatively as the horizon grows. Even small coefficient biases therefore translate into substantial distortions of the estimated responses at longer horizons (Kilian, 1998). In the limiting case of unit or near-unit roots, Phillips (1998) shows that long-horizon impulse responses from unrestricted VARs are not even consistently estimated, converging instead to random variables rather than the true responses. Second, the autoregressive coefficients themselves are biased downward in short samples. It is a long-established result that least squares underestimates the persistence of an autoregressive process (Marriott & Pope, 1954; Nicholls & Pope, 1988; Pope, 1990), and because $\hat{\Psi}_h$ inherits this bias through the recursion, the estimated impulse responses decay too quickly and understate the true persistence of the system. The distortion is most severe precisely when persistence is high and the sample is short. Third, these distortions undermine delta-method inference. When the largest roots approach the unit circle, the asymptotic normal approximation deteriorates and the resulting confidence intervals undercover, increasingly so at longer horizons (Kilian, 1998; Kilian, 1999).

Furthermore, a distinct class of distortions arises when the autoregressive structure is itself misspecified: if the true process has moving-average components, a finite-order VAR is generically inconsistent regardless of sample size, and while higher-order AR approximations can asymptotically absorb the omitted MA dynamics, the required lag expansion rapidly exhausts degrees of freedom in samples of typical macroeconomic length (Braun & Mittnik, 1993). The consequences of lag-order choice are strongly asymmetric—overfitting merely inflates variance, whereas underfitting eliminates higher-order dynamics of the response curve and produces spuriously tight intervals with coverage far below nominal levels (Kilian, 2001). Crucially, both channels are population-level phenomena: Ravenna (2007) shows that truncation bias in the VAR coefficients and the resulting contamination of the structural rotation matrix persist even as $T \to \infty$.

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2.2 Introduction to Local Projections

Whereas the VAR recovers the impulse response by iterating a single fitted model forward through powers of the companion matrix, the local projection estimates each horizon's response directly from a separate regression (Jordà, 2005). Following Jordà & Taylor (2025), the LP estimates the horizon-h response of the outcome y to the shock s_t directly, by projecting the future outcome y_{t+h} onto the current shock s_t while controlling for the conditioning set $\mathbf{x}_t$ that spans the information of the system,

$$y_{t+h} = \alpha_h + \beta_h \, s_t + \boldsymbol{\gamma}_h' \, \mathbf{x}_t + \xi_{t+h}^{(h)}, \qquad h = 0, 1, \ldots, H, \tag{12}$$

where y_{t+h} is the scalar response variable, and s_t the scalar shock of interest, which in the case of an observed structural innovation coincides with $s_t = \varepsilon_{j,t}$. The control vector $\mathbf{x}_t = (y_{t-1}', \ldots, y_{t-p}')'$ collects the p lags that span the conditioning information and under non-recursive identification may additionally absorb contemporaneous controls. Here α_h is a constant which can be ommitted without loss of generality, $\boldsymbol{\gamma}_h$ a $(Kp \times 1)$ vector of coefficients on the controls, and $\xi_{t+h}^{(h)}$ the horizon-h projection residual. The leading coefficient is the impulse response itself, β_h, which, when y is variable i and $s_t = \varepsilon_{j,t}$, coincides with the structural response θ_h^{ij} of Equation (6) and with the population object $\mathrm{IR}(h, \delta = 1)$ of Equation (1).

Under appropriate identification, in the simplest case as exogeneity of s_t conditional on $\mathbf{x}_t$, the impulse response is obtained directly as the OLS coefficient $\hat{\beta}_h$ on s_t in (12). If $s_t = \varepsilon_{j,t}$ is a unit-variance structural shock, $\hat{\beta}_h$ measures the response to a one-standard-deviation innovation. The LP thus recovers the impulse response without parametrizing the underlying propagation dynamics, in sharp contrast to the VAR's reliance on a fitted autoregressive model and forward iteration. When conditional exogeneity does not hold, causal interpretation can instead be enabled by enriching $\mathbf{x}_t$ to achieve selection on observables (Olea et al., 2025), by instrumenting s_t with an external proxy under a relevance and lead-lag exogeneity condition (Jordà et al., 2015; Stock & Watson, 2018) or by importing a conventional VAR identification scheme such as the recursive Cholesky ordering (Plagborg-Møller & Wolf, 2021). The instrumental-variable route is moreover robust to non-invertibility, recovering the response even when the structural shock is not spanned by current and past observables and the recursive scheme consequently fails.

Unlike the VAR, estimation of the structural impulse response $\hat{\beta}_h$ proceeds in a single stage, directly by OLS on Equation (12). This yields the h-step predictive regression of y_{t+h} on the shock of interest, whose leading coefficient coincides with the horizon-h impulse response (Jordà & Taylor, 2025). Since

$\hat{\beta}_h$ is the OLS coefficient on a regressor orthogonal to the projection error $\xi^{(h)}_{t+h}$ by construction of the structural shock, it recovers the impulse response directly, without propagation through a nonlinear mapping and hence without the delta-method step that governs the VAR estimator in Equation (11).

Provided that the process $\{y_t\}$ is strictly stationary and ergodic, with a positive definite covariance matrix, bounded moments $\mathbb{E}\|y_t\|^r < \infty$ for some $r > 4$, and mixing coefficients satisfying $\sum_{\ell=1}^{\infty} \alpha(\ell)^{1-4/r} < \infty$, the least-squares estimator is asymptotically normal (Hansen, 2022). Letting $n = T - h$ denote the effective sample size available at horizon h and $\mathbf{w}_t = (s_t, \mathbf{x}_t')'$ the stacked vector of all regressors, we have

$$\sqrt{n}\left(\hat{\mathbf{b}}_h - \mathbf{b}_h\right) \xrightarrow{d} \mathcal{N}(\mathbf{0},\ V_h)\,, \tag{13}$$

where $\mathbf{b}_h = (\beta_h, \boldsymbol{\gamma}_h')'$ is the vector of all projection coefficients and $\hat{\mathbf{b}}_h$ its OLS estimate. The asymptotic covariance matrix takes the robust sandwich form

$$V_h = Q^{-1}\Omega_h Q^{-1}, \tag{14}$$

where $Q = \mathbb{E}(\mathbf{w}_t\mathbf{w}_t')$ is the second-moment matrix of the regressors, and Ω_h is the long-run variance matrix of the projection scores, defined as

$$\Omega_h = \sum_{\ell=-\infty}^{\infty} \mathbb{E}\Big[\big(\xi^{(h)}_{t+h-\ell}\mathbf{w}_{t-\ell}\big)\big(\xi^{(h)}_{t+h}\mathbf{w}_t\big)'\Big]. \tag{15}$$

The long-run form of this covariance matrix reflects a fundamental structural feature of direct projection. Because the predictive regression aggregates shocks accruing between t and $t+h$ over overlapping forecast windows, the projection error $\xi^{(h)}_{t+h}$ inherently follows a moving-average process of order $h-1$. Consequently, the regression scores $\hat{\varepsilon}_{j,t}\,\xi^{(h)}_{t+h}$ are serially correlated, meaning the autocovariances in Ω_h do not vanish for $\ell \neq 0$.

To illustrate this mechanism, consider a stylized AR(1) model as the underlying DGP, with a persistence parameter ρ, initial condition $y_0 = 0$, and a strictly stationary error term u_t satisfying $\mathbb{E}[u_t|\{u_s\}_{s\neq t}] = 0$ almost surely (Jordà & Taylor, 2025):

$$y_t = \rho y_{t-1} + u_t. \tag{16}$$

Using a simplified version of the local projection estimator in (12), the h-step regression takes the form

$$y_{t+h} = \beta_h\, y_t + \xi^{(h)}_{t+h}. \tag{17}$$

Recursive substitution of the AR(1) law of motion into (17), using $\beta(\rho, h)$ to denote the LP IRF estimator for ρ^h and $\xi(\rho, h)$ the corresponding regression residual, yields

$$y_{t+h} = \beta(\rho, h)y_t + \xi(\rho, h), \qquad \xi(\rho, h) = \sum_{j=1}^{h} \rho^{h-j}\, u_{t+j}. \tag{18}$$

The stronger the persistence through ρ, the larger and more slowly decaying the autocovariances of $\xi(\rho, h)$, which exacerbates the distortion in naive OLS standard errors. Furthermore, as $\rho \to 1$, naive OLS estimation exhibits non-standard near-unit root behavior, invalidating inference based on standard normal critical values. However, as long as ρ lies safely inside the unit circle ($|\rho| < 1$), the LP estimator is not only consistent, yielding $\hat{\beta}_h \xrightarrow{p} \beta_h = \theta_h^{ij}$, but also remains asymptotically normal, provided the serially correlated regression residuals are appropriately corrected (Jordà & Taylor, 2025; Montiel Olea & Plagborg-Møller, 2021).

To address this serial correlation, Jordà (2005) originally proposed the use of Newey-West type (Newey & West, 1987) heteroskedasticity- and autocorrelation-consistent (HAC) standard errors. In practice, however, HAC inference introduces its own complications. It requires the choice of tuning parameters with complex interpretations (Lazarus et al., 2018) and can serve as an additional source of finite-sample bias. Specifically, Herbst & Johannsen (2024) show that while HAC estimators are asymptotically valid, estimates of the long-run variance are typically downward-biased in short samples, leading to a systematic understatement of the uncertainty surrounding the point estimate.

To tackle this and allow for valid inference even under near unity settings, Montiel Olea & Plagborg-Møller (2021) augment the local projection with one additional lag. Under the weak assumption that the shock is unpredictable from its own past and future values, this renders the regression score that is the product of the projection residual $\xi_{t+h}^{(h)}$ and the regressor of interest after partialling out the controls, serially uncorrelated, such that conventional Eicker–Huber–White (EHW) heteroskedasticity-robust standard errors suffice despite the serially correlated residual. The resulting t-statistic is asymptotically standard normal uniformly over $\rho \in [-1, 1]$ and horizons $h/T \to 0$.

A long-standing piece of conventional wisdom holds that local projections are more robust to model misspecification than VARs, originating with Jordà (2005, p. 162) and echoed across subsequent surveys (Li et al., 2024; Ramey, 2016). Plagborg-Møller & Wolf (2021) document this view while noting that it had long lacked formal foundation; in their finite-order treatment, LPs are indeed generally more robust to misspecification than equal-order VARs, a property theyestablish without invoking the population equivalence of the two estimands. Olea et al. (2024) show that, because the LP reads each horizon off a separate regression rather than iterating a single fitted model forward, a misspecification of the short-run dynamics does not compound through the powers of the companion matrix as it does for the VAR, whose bias arises precisely from extrapolating the response through the iterated mapping $\hat{A}^h$ that the misspecified model no longer satisfies.

However, similar to VAR models, Herbst & Johannsen (2024) show that LP point estimates are biased downward in finite samples for two reasons. First and distinct from VARs, the finite-sample estimation of the regression intercept introduces a non-zero distortion of the sample covariance between shock and outcome even when both are uncorrelated in population. Second and in analogy to VARs, the estimation of the lagged-variable coefficients inherits the classical downward bias of least-squares autoregressive estimators (Pope, 1990) and is amplified the more persistent the controls are. Both channels cause the bias to depend on the true impulse responses at neighbouring horizons and to worsen with the degree of persistence and the horizon, as the bias accumulates over an expanding sum of contributing terms as h increases. Although Herbst & Johannsen (2024) propose a bias corrected estimator, Li et al. (2024) find that it reduces bias only partially and at the cost of increased variance.

2.3 On the Equivalence of VARs and Local Projections

As local projection inference gained popularity over the last decade, a growing body of simulation evidence has compared the finite-sample performance of LP and VAR estimators across data-generating processes of differing richness. Two findings recur across the literature. When the DGP admits an exact finite-order VAR representation, the VAR estimator is correctly specified at the true lag order and dominates LP on mean-squared-error criteria at intermediate and long horizons and thus is always the preferable choice in terms of point estimation (Li et al., 2024; Olea et al., 2024). Furthermore in terms of coverage, Kilian & Kim (2011) and Brugnolini (2018) show that LP intervals are less accurate than an equivalent VAR benchmark. Even thought their stark findings largely rely on the underlying standard error estimation for local projections, Olea et al. (2024) do not observe an LP advantage when switching from HAC to EHW standard error estimation. Therefore the additional flexibility of least-squares LP yields little bias reduction relative to the already well-specified VAR, while carrying substantially higher variance.

When, however, the DGP departs from any finite-order VAR, as in the empirically calibrated DFM of Li et al. (2024) or via extionsion to VARMA processes, the underlying data does not admit an exact finite-order VAR representation and a genuine bias–variance trade-off emerges. In these settings the least-squares LP exhibits lower bias than the VAR estimator, but at the cost of substantially higher variance at intermediate and long horizons, so that VAR methods remain preferable under MSE unless researchers set an higher weigths on bias reduction Li et al. (2024). Furthermore, Olea et al. (2024) show that LPs using EHW inference yields correctly centered coverage intervals whereas VARs severly undercover while bias leads to incorrect centering. Also in cases of lag truncation, Brugnolini (2018) shows by varying the common lag length of LPs and VARs against a true VAR(12) process that LPs are able to increase their converage levels relative to VARs as the distance to the true lag length increases.

All of the aforementioned simulation studies compare LPs and VARs at a single common lag length p. This common-lag design, however, presupposes that the two methods estimate the same underlying object. Plagborg-Møller & Wolf (2021) establish that under weak stationarity, an unrestricted lag structure, and a common identification scheme, the LP and VAR estimators converge to the identical structural impulse response function in population, up to a constant of proportionality. Therefore, any conventional LP impulse response can be obtained through an appropriately ordered recursive VAR, and vice versa. Furthermore, because both methods act as best linear predictors of the underlying data-generating process, the authors show that they are symmetrically robust to non-linear misspecification in population.

Crucially for the equal-lag designs discussed above, Plagborg-Møller & Wolf (2021) further demonstrate that when both specifications are restricted to a fixed order p, their population estimands continue to agree exactly up to horizon $h \leq p$, but generally diverge for $h > p$. This theoretically localizes the divergence to horizons beyond the estimation lag length, thus precisely the intermediate and long horizons where the simulation literature typically finds the bias-variance trade-off to be most pronounced. Consequently, the choice between LP and VAR in practice is not a choice between fundamentally different population objects, but rather between two finite-sample estimators of the same structural response. What distinguishes them is primarily their bias and variance in finite samples.

While Plagborg-Møller & Wolf (2021) locate the theoretical divergence at horizons $h > p$, they do not analytically characterize the exact implicit extrapolation the local projection performs in finite samples. Ludwig (2026) closes this gap by establishing an exact least-squares identity that holds not just in population, but in any given finite sample. Specifically, the h-step impulse response of an LP(p) is algebraically equivalent to the forward iteration of h one-step predictions from a sequence of VARs of increasing order:

$$\mathrm{IRF}_h^{\mathrm{LP}(p)}(d) = \begin{bmatrix} I_K & \mathbf{0} \end{bmatrix} \Pi(p+h-1) \cdots \Pi(p+1)\, \Pi(p) \begin{bmatrix} d \\ \mathbf{0} \end{bmatrix}, \tag{19}$$

where $\Pi(q)$ denotes the estimated companion matrix of a VAR(q), I_K is the identity matrix matching the dimension of the system, and d represents the initial structural shock vector.

At $h = 1$, the two methods coincide identically. However, at $h = 2$, the LP(p) is no longer the simple two-step iterate of a VAR(p). Instead, it advances the one-step forecast using a VAR($p+1$) and each further horizon effectively adds one lag to the implicit VAR representation. This identity inherits a profound implications for model comparison. At equal-order LPs and VARs are not equally parsimonious or expressed in the same recursive language, an LP(p) at horizon h is constructed from VARs of orders $p, p+1, \ldots, p+h-1$, whereas a standard VAR(p) holds the order rigorously fixed at every step.

The fixed-order VAR(p) is therefore a heavily restricted special case of the equal-order LP(p), which in turn is a restricted special case of a highly parameterized VAR($p+h-1$). From this perspective, the familiar empirical pattern of lower approximation bias but higher sampling variance for LPs relative to a same-order VAR is largely a mechanical consequence of comparing a more flexible, over-parameterized specification with its own restriction, rather than an intrinsic property of the projection method itself. This fundamentally confounds the standard reading of the bias-variance trade-off as at equal lag orders, the methodological effect (LP vs. VAR) cannot be cleanly separated from the complexity effect (more vs. fewer effective lags).

3. Baseline Simulation

This confounding motivates the first part of our study. Because Ludwig (2026)'s contribution is analytical and deliberately abstains from a simulation-based horse race, the question of whether the conventional finite-sample conclusions survive once the equal-lag restriction is relaxed remains open. We therefore operationalize Ludwig's central implication through a complexity-adjusted benchmarking exercise that holds effective model complexity constant, comparing a fixed LP(4) estimator against progressively higher-order VAR specifications that proxy its increasing dynamic flexibility. Crucially, because the identity in Theorem 1 maps LP(p) to a sequence of VARs rather than to any single higher-order VAR, no individual VAR is the perfect counterpart. Instead, LP(4) is read as a reference line against the full VAR-order frontier, against which its bias, variance, and coverage can be accurately located.

3.1 Simulation Design

The baseline study serves two purposes. First, it operationalizes the complexity argument of Ludwig (2026) in a controlled, correctly specified setting. Rather than comparing LP(4) to a single equal-lag VAR(4), we locate the fixed LP(4) estimator on the full VAR-order frontier, ranging from VAR(1) to VAR(p+H-1), thereby rendering the implicit complexity of the direct projection visible as a position on that frontier. Second, it establishes the correctly specified reference design against which later extensions measure the effect of dynamic and functional misspecification.

The underlying DGP is a stable bivariate zero-mean VAR(4) with recursive Cholesky identification. Consistent with the theoretical setup, this imposes a contemporaneous zero restriction where the first variable does not respond on impact to shocks in the second variable, while the second variable may respond to both structural shocks simultaneously.

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + u_t, \quad y_t \in \mathbb{R}^2 \tag{20}$$

$$\Sigma_u = BB', \qquad B = \begin{pmatrix} 1 & 0 \\ 0.1 & 0.5 \end{pmatrix}, \quad b_{11}, b_{22} > 0 \tag{21}$$

$$\rho(\mathcal{A}) < 1, \qquad \mathcal{A} = \begin{pmatrix} A_1 & A_2 & A_3 & A_4 \\ I_2 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & I_2 & \mathbf{0} \end{pmatrix} \in \mathbb{R}^{8\times 8} \tag{22}$$

Persistence is calibrated via the spectral radius of the companion matrix $\mathcal{A}$ across the empirically relevant range $\rho \in \{0.5, 0.7, 0.95\}$ (Figure 1). The lower bound reflects weakly persistent processes where impulse responses decay rapidly, rendering bias-variance comparisons at longer horizons uninformative. The upper bound approaches the unit-root boundary while maintaining covariance stationarity, covering the high-persistence region where finite-sample differences between estimators are most pronounced.

Sample sizes are set to $T \in \{100, 250, 500\}$ observations, generated after a burn-in period of 200 periods to eliminate initialization effects. The choice of $T = 100$ aligns precisely with the median sample size documented by Herbst & Johannsen (2024) in the applied structural VAR literature. At the same time, this value serves as the effective lower bound for our specific simulation design. Because our complexity-adjusted benchmarking requires estimating highly parameterized models up to VAR($p+H-1$), pushing the sample size below this median would rapidly exhaust degrees of freedom and yield computationally unstable estimates. The intermediate value $T = 250$ is therefore included to represent a comfortably large empirical dataset, corresponding to twenty years of monthly or sixty years of quarterly observations. This ensures reliable estimation across the entire VAR-order frontier. Finally, the value $T = 500$ serves as a large-sample reference point to assess convergence toward the population equivalence established by Plagborg-Møller & Wolf (2021).

The structural estimand is $\theta_h = \Psi_h B e_1$, where Ψ_h denotes the h-step ahead moving-average matrix of the reduced-form VAR, generated via the recursion $\Psi_h = \sum_{j=1}^{\min(h,4)} A_j \Psi_{h-j}$ with $\Psi_0 = I_2$, and $e_1 = (1, 0)'$ as the selection vector. The primitive shock is normalized as a unit shock. We trace the dynamic response of the system to a one-standard-deviation increase in the first structural innovation $\varepsilon_{1,t}$.

$$u_t = B\varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, I_2) \tag{23}$$

The innovations are specified as Gaussian white noise. This standard assumption acts primarily as an expositional device, allowing conditional expectations to replace linear projections (Plagborg-Møller & Wolf, 2021). Because our performance metrics (bias and variance) depend strictly on the second moments of the data, the exact distribution of the innovations is irrelevant for the baseline estimator comparison. Furthermore, holding the Gaussian innovation structure fixed isolates the effects of the structural misspecifications introduced later in the study ensuring differences in performance are driven solely by the specified channel.

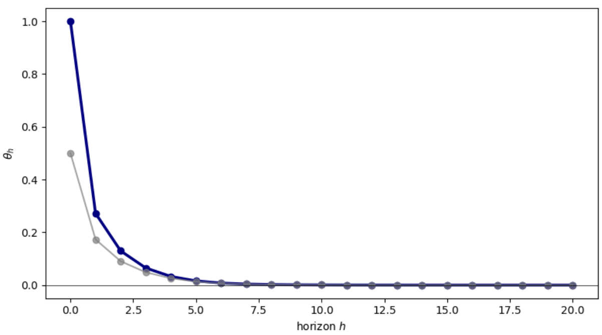
The structural target of interest is the impulse response function, which tracks the dynamic causal effect of a structural innovation on the variables of the system over time. Formally, the population estimand at horizon h is defined as the vector of responses to a one-standard-deviation shock in the first structural innovation $\varepsilon_{1,t}$. This object is constructed analytically from the data-generating process parameters as

$$\theta_h = \Psi_h B e_1, \qquad e_1 = (1,0)', \qquad h = 1,\ldots,20 \tag{24}$$

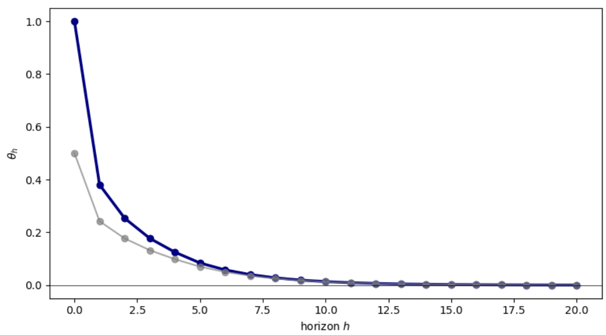
where Ψ_h represents the reduced-form moving-average coefficient matrix obtained from the companion recursion, B is the lower-triangular Cholesky factor of the innovation variance matrix, and e_1 serves as the selection vector. The multiplication by e_1 isolates the first column of the structural impact matrix, which restricts the analysis to the propagation of the first shock while discarding contemporaneous variations from the second structural innovation. Because the baseline data-generating process is linear and covariance stationary, this population estimand represents the true conditional expectation of the system given the structural shock. It constitutes the invariant benchmark that both the iterated VAR(q) specifications and the direct LP(4) regressions attempt to recover in finite samples.

Figure 1: True Impulse Response Functions Across Persistence Regimes

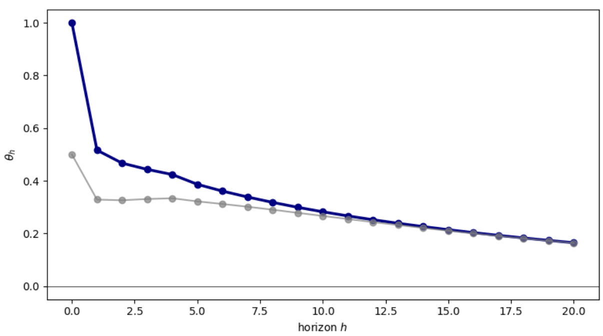
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



Note. Each panel reports the true structural impulse response function of variables one and two to a one-standard-deviation shock in the first structural innovation, traced over horizons $h = 1,\ldots,20$. Panels correspond to persistence regimes $\rho \in \{0.5, 0.7, 0.95\}$. Blue line represents $y_1 \leftarrow \epsilon_1$(*estimand* θ_h), while grey line shows $y_2 \leftarrow \epsilon_1$.

We compare two classes of models. The first is a fixed LP(4) estimator, which serves as the reference point. At each horizon $h = 1, \ldots, H$, the LP(4) estimator is obtained from the OLS regression

$$y_{1,t+h} = \mu_h + \beta_h \hat{\varepsilon}_{1,t} + \sum_{\ell=1}^{4} \delta'_{h,\ell} y_{t-\ell} + \xi_{h,t}, \quad h = 1, \ldots, 20 \tag{25}$$

where $\hat{\varepsilon}_{1,t} = e'_1(\hat{B}^{(4)})^{-1}\hat{u}_t^{(4)}$ is the estimated structural shock recovered from a first-stage VAR(4). While the true structural shock is perfectly observable by construction in our data-generating process, directly projecting on this true innovation would artificially advantage the local projection by bypassing identification uncertainty. To maintain a rigorously fair comparison, we intentionally restrict the LP(4) to use the empirically estimated shock. This guarantees that both estimators face the exact same identification problem. Consequently, any divergence in finite-sample performance is driven strictly by differences in dynamic propagation rather than discrepancies in shock recovery. The structural impulse response estimate at horizon h is then simply the slope coefficient $\hat{\theta}_h^{\text{LP}} = \hat{\beta}_h$.

The second class consists of a sweep of VAR estimators of increasing order, VAR(q) for $q = 1, \ldots, p + H - 1$, where $p = 4$ and $H = 20$. For each order q, the VAR is estimated equation-by-equation via OLS. The reduced-form covariance matrix $\hat{\Sigma}_u^{(q)}$ is Cholesky-decomposed to recover the structural impact matrix $\hat{B}^{(q)}$, and the h-step ahead structural IRF is computed as

$$\hat{\theta}_h^{\text{VAR}(q)} = \hat{\Psi}_h^{(q)} \hat{B}^{(q)} e_1 \tag{26}$$

where $\hat{\Psi}_h^{(q)}$ denotes the h-step ahead moving-average matrix of the estimated VAR(q).

This sweep operationalizes the complexity-adjusted benchmark. By Theorem 1 of Ludwig (2026), LP(4) at horizon h is algebraically equivalent to a sequence of VAR predictions stepping through orders $4, 5, \ldots, 4 + h - 1$. The rigorous comparison is therefore not LP(4) against VAR(4) alone, but against the full range of VAR specifications up to order $p + H - 1$. Finally, no data-dependent lag selection is performed within the sweep. All VAR orders $q = 1, \ldots, p + H - 1$ are systematically estimated on each simulated sample to guarantee that the comparison evaluates fixed specifications rather than selection procedures.

Performance is assessed along two dimensions and reported horizon-by-horizon for $h = 1, \ldots, 20$. For point estimation, we evaluate the bias, variance, mean squared error, and root mean squared error of the impulse response estimates relative to the true population estimand. These metrics are computed across $B = 5{,}000$ Monte Carlo replications, ensuring tight simulation bounds in line with recent literature (Li et al., 2024). To explicitly quantify this simulation uncertainty, Monte Carlo standard errors are calculated and reported alongside all point-estimator performance measures. Formal mathematical definitions of these sample estimators are deferred to the Appendix.

For inference, we assess the empirical coverage of nominal 95% confidence intervals, defined strictly as the fraction of replications in which the estimated interval successfully brackets the true structural estimand. For the VAR(q) sweep, confidence intervals are constructed via the delta method, which analytically propagates the estimation uncertainty of the reduced-form coefficients through the nonlinear mapping to the structural impulse responses (Kilian & Lütkepohl, 2017). For the fixed LP(4) estimator, confidence intervals are based on heteroskedasticity-robust standard errors. This approach is asymptotically valid and sufficient given the martingale difference assumption on the structural shock under recursive identification (Plagborg-Møller & Wolf, 2021).

3.2 Simulation Results

Figure 2 introduces the complexity frontier, defined here as the horizon-averaged root mean squared error of the recursively identified VAR impulse response estimator plotted as a function of the estimation lag order q. The frontier is traced across the full sweep from $q = 1$ to $q = 23$ for the three persistence regimes $\rho \in \{0.5, 0.7, 0.95\}$ and the sample sizes $T \in \{100, 250, 500\}$. The fixed LP(4) estimator serves as a horizontal reference line for each respective sample size.

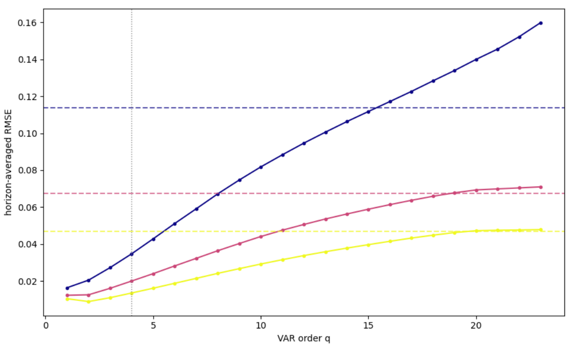
Two overarching patterns govern the general level of the root mean squared error across the panels. First, an increase in the sample size yields a strict monotonic decline in the error metric. Within every persistence regime, the curve and the corresponding LP(4) reference line for $T = 500$ lie completely below those for $T = 250$, which in turn lie strictly below the values for $T = 100$. Second, an increase in the persistence parameter ρ systematically elevates the overall level of the errors. Comparing the panels from left to right reveals that the entire complexity frontier, along with the horizontal reference lines, shifts strictly upward as the system becomes more persistent.

Focusing on the conventional equal-lag comparison at $q = 4$, the solid VAR curves lie strictly below the dashed LP(4) lines of the corresponding color in all panels and for all sample sizes. When relaxing the equal-lag restriction and tracing the full frontier, the shape of the VAR curves varies noticeably across the persistence regimes. At $\rho = 0.5$, the curves appear smooth and, after passing a shallow minimum near $q = 2$, increase monotonically. At $\rho = 0.7$, a mild kink emerges in the early portion of the curve, located before the four-lag mark. At the high persistence level of $\rho = 0.95$, this initial feature becomes highly pronounced, with the curves dropping steeply to a distinct minimum around $q = 2$ to $q = 3$ before rising again to form a clear U-shape.

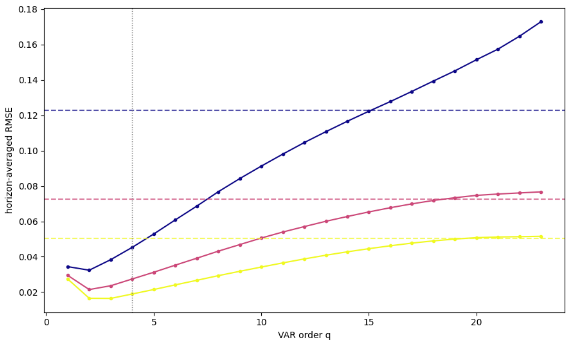
Furthermore, across all persistence regimes, the shape of the VAR curves changes systematically with the sample size. As T increases, the curves become noticeably flatter and their minimum shifts to the right. This indicates that in larger samples, the lowest error is achieved at a higher lag order, and the curves rise much more gradually thereafter. Because of this flattening effect, the point where the rising VAR frontier eventually intersects the horizontal LP(4) benchmark also moves further to the right as T grows. This intersection is most distinctly observable in the smallest sample size $T = 100$, where the curves are steepest. At $\rho = 0.5$ and $\rho = 0.7$, the LP(4) line intersects the $T = 100$ frontier roughly in the neighborhood of $q \approx 15$. Under the high persistence regime of $\rho = 0.95$, this intersection point shifts leftward to a slightly lower VAR order.

Figure 2: Complexity Frontier Across Persistence Regimes

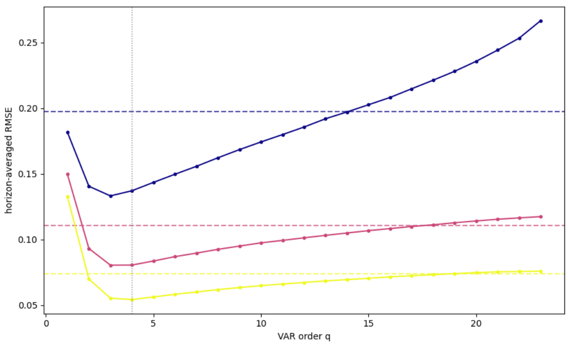
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



Note. Horizon-averaged RMSE of the structural IRF estimate as a function of VAR order $q = 1, \ldots, 23$ (solid lines), with the fixed LP(4) estimator shown as a horizontal reference line of the same colour (dashed), for each sample size $T \in \{100, 250, 500\}$. The vertical dotted line marks the equal-lag VAR(4); B = 5,000 replications. Panels correspond to persistence $\rho \in \{0.5, 0.7, 0.95\}$. Colors blue, red, yellow represent sample sizes 100, 250, 500, respectively.

Figure 3 plots the point estimation performance in terms of bias, variance, and root mean squared error for a fixed sample size of $T = 250$. The results are reported horizon by horizon and grouped by the three persistence regimes. The fixed LP(4) estimator is compared against a representative selection of VAR specifications of varying lag orders. Across all persistence regimes the specific horizon at which the variance paths of the different estimators begin to diverge from one another remains largely consistent, increasing slightly with persistence.

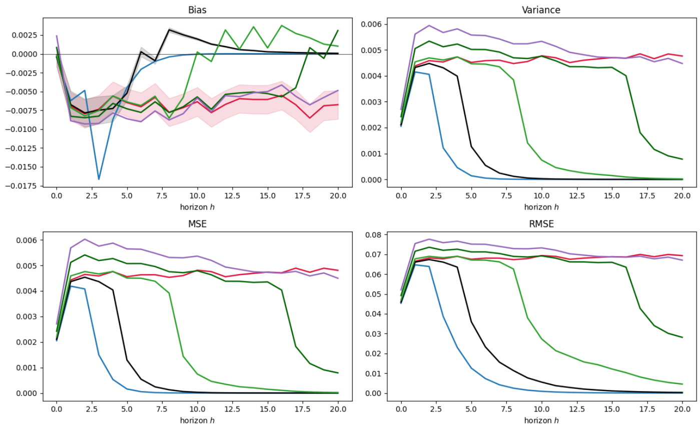
Under the low persistence calibration ($\rho = 0.5$), the absolute bias across all classes grows until the second to third horizon before splitting into distinct paths. The under-parameterized VAR(2) specification appears erratic at short horizons but, together with VAR(4), returns to unbiasedness the fastest around horizon eight. Increasing the VAR order causes the bias paths to initially track the LP(4) reference line before eventually shrinking toward zero at intermediate horizons, whereas the VAR(23) and the LP(4) coincide completely. The variance, and consequently the root mean squared error, paints a similar picture. The parsimonious models VAR(2) through VAR(8) track the LP(4) curve until horizon two before falling back toward zero, with VAR(2) declining the fastest. The heavily parameterized VAR(16) and VAR(23) models overshoot the local projection variance at first but decline significantly at intermediate and long horizons. Meanwhile, the variance of the LP(4) estimator remains comparatively stable across the entire horizon.

At the intermediate persistence level ($\rho = 0.7$), the bias exhibits a similar structural pattern but increases noticeably in absolute terms. The VAR(2) specification now overshoots completely, leaving the correctly specified VAR(4) as the most accurate model. The intermediate VAR(8) and VAR(16) models track the LP(4) trajectory for a substantially longer period before their bias dissipates, and VAR(23) again closely mirrors the local projection across the full horizon. The variance experiences only a minuscule increase compared to the low persistence regime. The main difference is that the decline from the early peak is now stretched over a longer horizon. Because the bias remains relatively small for most specifications, the root mean squared error closely resembles the variance pattern.

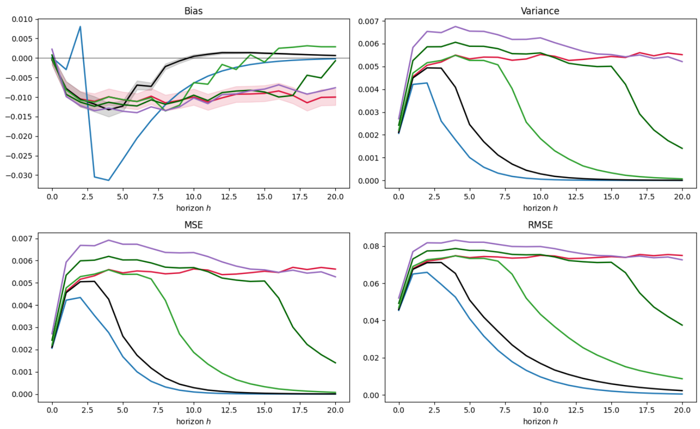
Near the unit root boundary ($\rho = 0.95$), the point estimation landscape changes considerably. Absolute bias is highly elevated across all specifications, with VAR(2) behaving in a highly erratic manner. The remaining VAR specifications and the local projection share a similar downward trajectory and roughly track each other across the full horizon. Variance is similarly elevated across the board. The VAR(4) model achieves the lowest variance, followed closely by VAR(2). Unlike in the less persistent regimes, none of the specifications regain efficiency at longer horizons, leaving the dispersion elevated and producing a distinctly arched profile for the root mean squared error.

Figure 3: Point-Estimation Performance Across Persistence Regimes

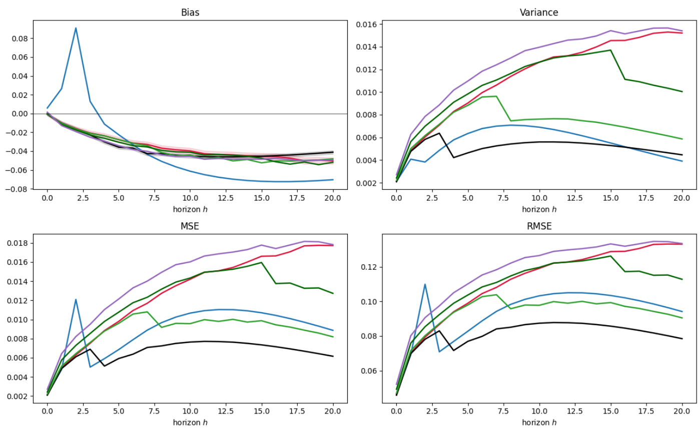
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



Note. Each panel reports bias, variance, and RMSE of the estimated impulse responses $\hat{\theta}_h$ relative to the true estimand θ_h, horizon-by-horizon for $h = 1, \ldots, 20$, computed across $B = 5{,}000$ Monte Carlo replications with $T = 250$. The fixed LP(4) reference line is compared against the complexity-adjusted VAR(q) sweep. Panels correspond to persistence regimes $\rho \in \{0.5, 0.7, 0.95\}$. Shaded area in bias represents 95% band from Monte Carlo standard errors. Line color represents model architecture, with blue, black, light green, dark green, lilac, red, representing VAR(2), VAR(4), VAR(8), VAR(16), VAR(23), and LP(4), respectively.

Figure 4 displays the empirical coverage of nominal 95 percent confidence intervals in the left column and compares the VAR(4) delta-method standard error against the empirical sampling standard deviation in the right column. The results are plotted horizon by horizon across the three persistence regimes for a sample size of $T = 250$.

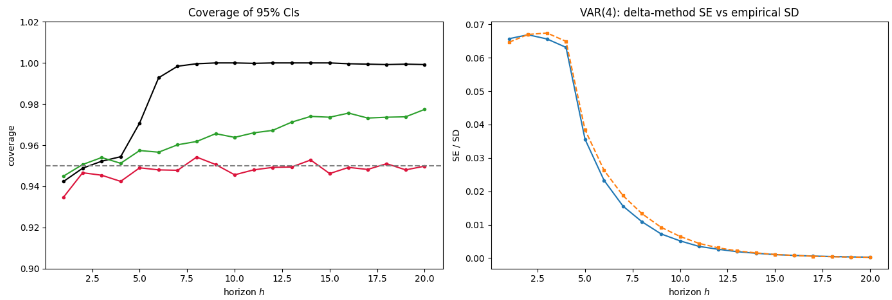
Under the low persistence calibration of $\rho = 0.5$, all models achieve roughly the same coverage level up to the third horizon. A visible divergence begins at the fourth horizon. The coverage of the VAR(4) specification increases rapidly and reaches full coverage by the seventh horizon. The over-parameterized VAR(23) model slowly converges toward a coverage of 1.0 by the end of the forecast window. Meanwhile, the LP(4) estimator remains highly consistent, fluctuating tightly around the 0.95 level throughout the entire horizon. In the right panel, the delta-method standard error closely tracks the empirical standard deviation, showing only a marginal deviation around the initial peak.

At the intermediate persistence level of $\rho = 0.7$, the coverage paths of all estimators remain tightly aligned until the sixth horizon. Beyond this point, the VAR(4) coverage steadily deteriorates toward the end of the observed window. The VAR(23) specification exhibits a slight upward drift in coverage, whereas the LP(4) estimator maintains a stable trajectory near the nominal target. The corresponding standard error validation in the right column shows that the analytical standard error and the empirical standard deviation lie almost perfectly on top of each other.

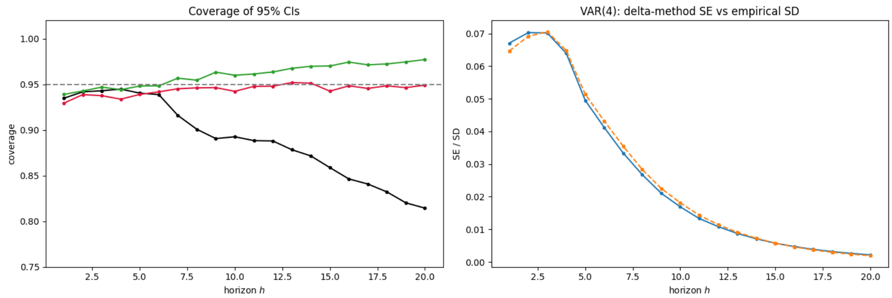
Approaching the unit root boundary at $\rho = 0.95$, the coverage behavior changes substantially. The models track each other only until the second horizon, after which none of the estimators reach the nominal 95 percent coverage target. The VAR(4) specification experiences a severe deterioration, with its coverage dropping below the 0.8 mark at longer horizons. The VAR(23) model performs slightly better and closely tracks the local projection, yielding a coverage level of roughly 0.9 across the remainder of the horizon. In the right column, a distinct vertical gap emerges. The empirical sampling dispersion shifts visibly above the analytical delta-method standard error at short and intermediate horizons, before the two curves gradually converge toward the end of the forecast window.

Figure 4: Confidence-Interval Coverage Across Persistence Regimes

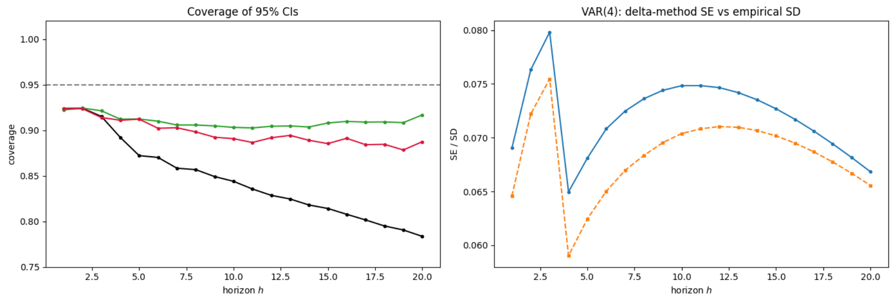
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



Note. Each subfigure corresponds to a persistence regime $\rho \in \{0.5, 0.7, 0.95\}$ and contains two panels. The *left panel* reports the empirical coverage of nominal 95% confidence intervals — the fraction of replications in which the interval contains the true estimand θ_h — horizon-by-horizon for $h = 1, \ldots, 20$, with $T = 250$ and $B = 5{,}000$. Coverage is shown for VAR(4) (black), VAR(23) (green), and LP(4) (red), with the dotted horizontal line marking the nominal 95% level. VAR(q) intervals are constructed via the delta method; LP(4) intervals use heteroskedasticity-robust standard errors. The *right panel* validates the VAR(4) delta-method standard error against the empirical Monte Carlo sampling standard deviation of $\hat{\theta}_h$: the empirical SD is shown in blue and the mean delta-method SE in orange, so that close agreement indicates an accurate analytical SE while a gap signals that it mis-states the true sampling dispersion.

3.3 Discussion of Results

The complexity frontier operationalizes the central implication of Ludwig (2026). Because an LP(p) estimation algebraically embeds a sequence of VAR predictions from orders p through $p + h - 1$, the mathematically coherent benchmark for a local projection is not a single equal-order VAR but the entire VAR-order sweep. The observed baseline efficiency advantage of the equal-lag VAR(4) strictly replicates the standard consensus in the literature (Li et al., 2024; Olea et al., 2025). However, locating the local projection on the frontier reveals the profound asymmetry of this conventional comparison. The local projection does not sit near the parsimonious VAR specification. Instead, it inherently operates at the complexity level of a highly parameterized system, intersecting the sweep at orders higher than $q \approx 15$. This confirms empirically that a sufficiently augmented VAR behaves identically to a local projection. The standard finite-sample superiority of the VAR therefore reflects a disparity in effective model complexity rather than an intrinsic deficiency of the direct projection method.

Beyond model complexity, the underlying bias-variance tradeoff is heavily dictated by the persistence of the data-generating process. The monotonic frontier under low persistence illustrates that the variance penalty for over-parameterization heavily outweighs any reduction in bias when the true impulse response decays rapidly. Conversely, the pronounced U-shape approaching the unit root boundary highlights the severe cost of lag truncation. Under near-integrated dynamics, the downward bias affecting autoregressive coefficients dominates the error budget (Kilian, 1998). Crucially, the local projection loses its theoretical robustness advantage here. Because it provides no relief from this persistence-driven bias while simultaneously incurring the inherently higher variance of its direct horizon-by-horizon regressions, the local projection is strictly dominated by the equal-lag VAR. The parametric estimator wins near the non-stationary boundary simply by virtue of its smaller variance footprint. This demonstrates that while the finite-sample superiority of the VAR survives the complexity adjustment, its magnitude is strictly an artifact of the data-generating process.

Evaluating performance solely through root mean squared error obscures a critical theoretical divergence between point estimation and inference. The coverage results expose that the parametric efficiency of the VAR comes at the cost of inferential fragility. The delta-method confidence intervals of the VAR(4) model fail for structural reasons across the persistence spectrum. At weak persistence, the intervals artificially over-cover because the underlying response converges to zero, causing the interval to mechanically capture the unconditional mean rather than indicating high statistical precision. At high persistence, the first-order Taylor approximation underlying the delta method breaks down. It severely understates the true empirical dispersion of the skewed finite-sample distribution (Kilian, 1999), leading to steep under-coverage. In contrast, the LP(4) estimator maintains near-nominal coverage across all regimes. This empirically validates the double robustness of local projections (Montiel Olea & Plagborg-Møller, 2021), showing that heteroskedasticity-robust standard errors remain valid even when the dynamic process approaches non-stationarity. A superior estimand efficiency therefore does not guarantee valid inference.

Our baseline findings are subject to two notable methodological limitations. First, the complexity-adjusted logic strictly requires estimating highly parameterized models up to VAR(23). Pushing the sample size below the applied median of $T = 100$ renders these high-order systems rank-deficient and computationally unstable. Consequently, we are unable to explore the simulation space where the finite-sample bias-variance tradeoff is most aggressive and where the robustness arguments for local projections might be most practically relevant. Second, the complexity frontier collapses bias and variance into a single root mean squared error metric aggregated over the forecast horizon. While this efficiently visualizes the consistency effect as the sample size grows, it obscures the exact mechanism driving the improvement. Because the explicit decomposition into bias and variance is restricted to a fixed sample size, the question of whether the shift in the frontier across sample sizes is predominantly bias-driven or variance-driven remains partially unresolved.

4. Extension 2: Functional Misspecification (3-4 pages)

4.1 Simulation Design

Extension 2 examines whether the finite-sample MSE ranking between LP(4) and the complexity-adjusted VAR sweep is robust to functional misspecification of the DGP. The baseline VAR(4) is augmented by a mean-zero quadratic term in the first lag, yielding a nonlinear DGP:

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + \gamma \cdot g(y_{1,t-1}) + u_t, \quad u_t = B\varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I_2) \tag{27}$$

where

$$g(y_{1,t-1}) = \left(0,\ \left(\frac{y_{1,t-1}}{\sigma_1}\right)^2 - 1\right)' \tag{28}$$

is a standardised, mean-zero quadratic in the first variable's first lag, entering the second equation only, with $\sigma_1 = \sqrt{\mathrm{Var}(y_{1,t})}$ the stationary standard deviation of $y_{1,t}$ under the linear ($\gamma = 0$) VAR, and $\gamma \in \{0, 0.3, 0.6\}$ controlling the degree of nonlinearity.

Importantly, adding a quadratic term in lagged levels can increase the risk of non-stationary or explosive simulated paths, particularly under high shock persistence. For this reason, this extension restricts the persistence parameters to $\rho \in \{0.5, 0.7\}$. This restriction is intended to mitigate violations of covariance stationarity and to keep the estimand θ_h well-defined over the simulated designs. This specification changes the nature of the misspecification relative to the baseline linear VAR(4). The nonlinear term enters the propagation dynamics through a quadratic function of $y_{1,t-1}$, while the structural shocks continue to enter contemporaneously and linearly through $u_t = B\varepsilon_t$. Thus, the exercise does not isolate nonlinear shock transmission, but studies whether the finite-sample MSE ranking between LP(4) and the complexity- adjusted VAR sweep is robust when the true conditional mean is nonlinear in lagged state variables. Because the nonlinear component is a function of lagged outcomes, increasing the lag length of a linear VAR cannot in general eliminate the functional misspecification and the fitted models remain linear projections of a nonlinear data-generating process.

The key theoretical motivation for this extension follows from Plagborg-Møller & Wolf (2021), who establish that linear LP and linear VAR estimate the same best linear approximation to the true IRF in population, regardless of whether the DGP is linear (Proposition 1, p. 960). Under comparable lag information, both estimators thus provide an equal linear approximation of the non-linear DGP, and any finite-sample divergence reflects differences in the bias-variance tradeoff rather than differential robustness to nonlinearity per se. This makes Extension 2 complementary to Extension 1, as whereas Extension 1 isolates the bias channel under invertible dynamic misspecification, Extension 2 isolates functional-form misspecification in the conditional mean.

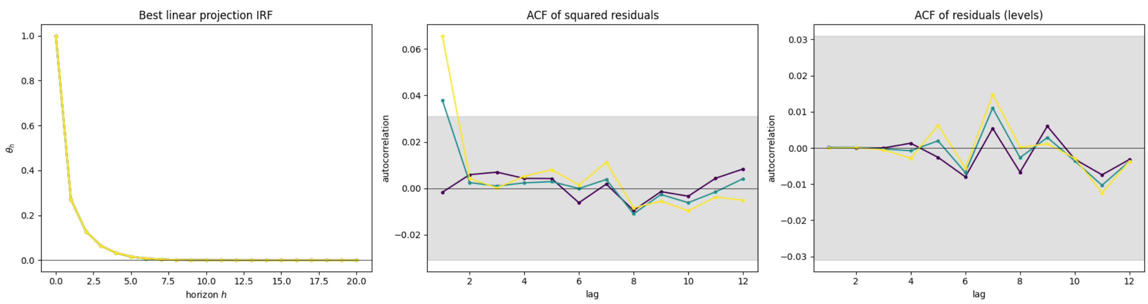
The structural estimand is the best linear approximation to the impulse response generated by the nonlinear DGP, denoted $\theta_h^{\text{lin}} = \Psi_h^{\text{lin}} B^{\text{lin}} e_1$. Here Ψ_h^{lin} denotes the linear projection response implied by the covariance structure of the nonlinear process, and B^{lin} is the corresponding impact matrix obtained from the reduced-form innovation covariance matrix using the same recursive identification scheme as in the baseline design. Since both LP and VAR target the same linear projection estimand in population under the conditions described by Plagborg-Møller & Wolf (2021), the comparison remains well-defined under nonlinear misspecification.

Persistence is varied across $\rho \in \{0.5, 0.7\}$, while sample sizes are chosen to reflect small sample and asymptotic behaviors using $T \in \{240, 10000\}$. The nonlinearity strength $\gamma \in \{0.3, 0.6\}$ is varied jointly with persistence and sample size. Both estimators are applied without modification to data generated by the nonlinear DGP, so the functional-form misspecification is fully reflected in the finite-sample performance of each estimator. Performance measures are identical to the baseline, with added small-sample and asymptotic analysis of deviation from the ground truth IRF, to investigate the claim of equivalence in population.

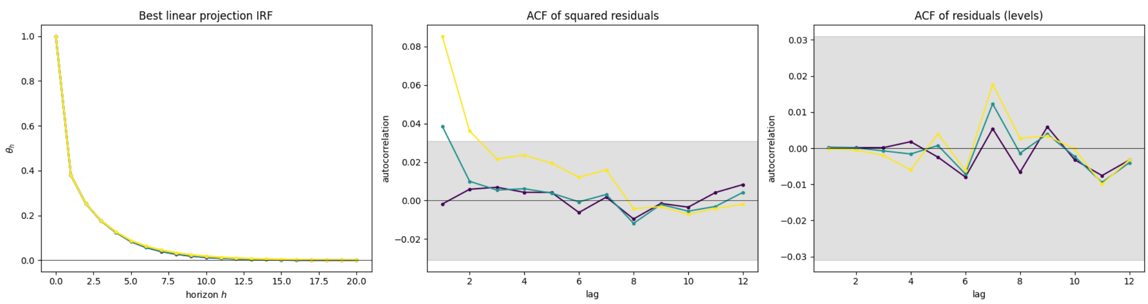
4.2 Simulation Results

Figure 5: True IRF Functions by Nonlinearity Strength

(a) $\rho = 0.5$



(b) $\rho = 0.7$

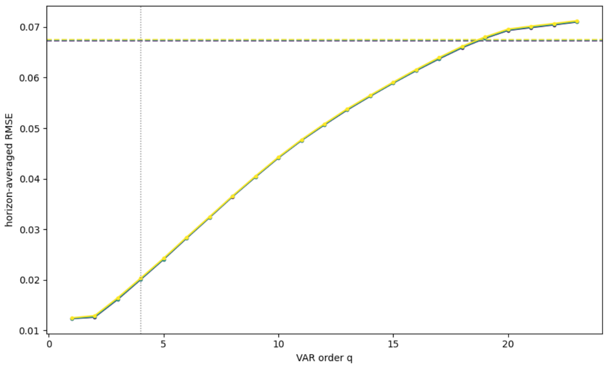


Note. Each subfigure ($\rho \in \{0.5, 0.7\}$) has three panels. *Left:* the best linear-projection IRF θ_h of variable 1 to shock 1, approximately invariant across nonlinearity strength γ. *Middle, right:* the ACF of the squared and level second-equation residual ($\hat{e}_{2,t}^2$, $\hat{e}_{2,t}$). The squared-residual ACF lifts above the band as γ grows (conditional heteroskedasticity), while the level residual stays white. Lines: purple $\gamma = 0$ (baseline), teal $\gamma = 0.3$, yellow $\gamma = 0.6$; the grey area is the $\pm 1.96/\sqrt{N}$ 95% white-noise band.

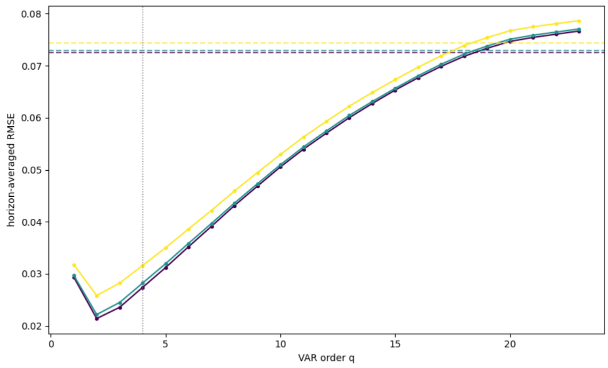
Figure 5 shows the true IRF together with the autocorrelation of the squared and level second-equation residuals of a VAR(4). As γ and ρ increase, the squared residuals develop significant autocorrelation at lag 1, while the level residuals stay within the white-noise band. The nonlinear term is thus approximately orthogonal to the linear lag space and leaves the best linear projection essentially unchanged. The misspecification only surfaces as conditional heteroskedasticity in the residuals. Furthermore, because the delta-method standard errors assume homoskedasticity, they are invalid under this residual heteroskedasticity. This leads to VAR(4)'s confidence intervals under-covering, whereas LP's heteroskedasticity-robust intervals remain valid.

Figure 6: Complexity Frontier under Nonlinear Lag Misspecification

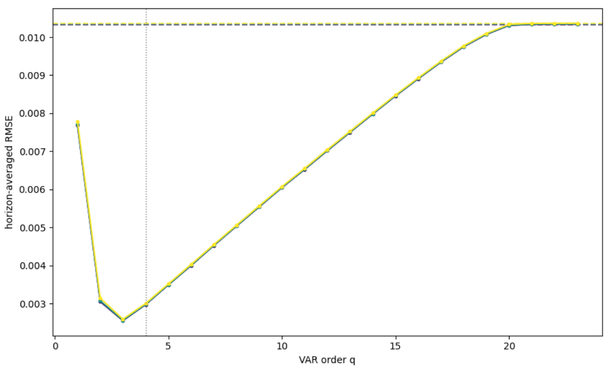
(a) $\rho = 0.5$ (Sample)



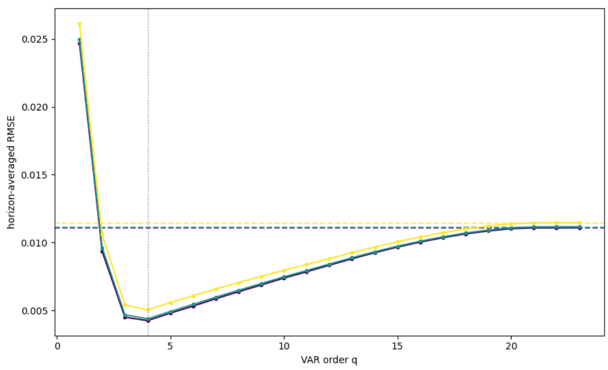
(b) $\rho = 0.7$ (Sample)



(c) $\rho = 0.5$ (Asymptotic)



(d) $\rho = 0.7$ (Asymptotic)



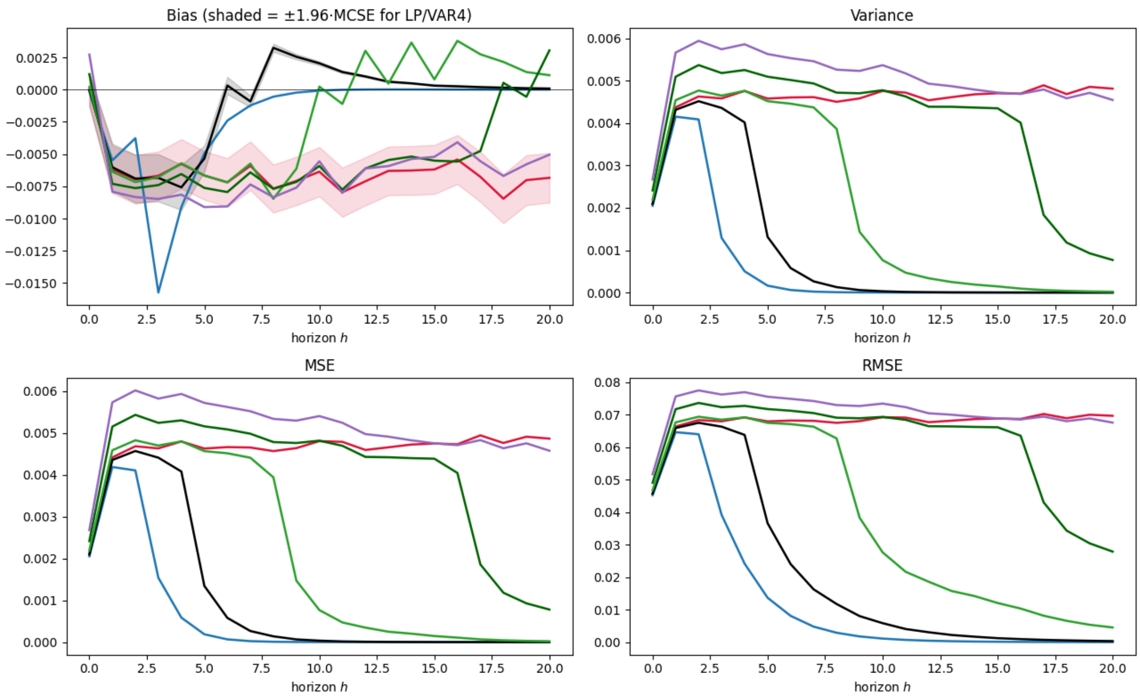
Note. Horizon-averaged RMSE of the structural IRF against VAR order $q = 1, \ldots, 23$ (solid), with LP(4) a same-colour horizontal dashed reference and the vertical dotted line marking the equal-lag VAR(4). One colour per nonlinearity strength: purple $\gamma = 0$ (baseline), teal $\gamma = 0.3$, yellow $\gamma = 0.6$. Top row (a, b) is the small-sample frontier $T = 250$, bottom row (c, d) the asymptotic frontier $T = 10{,}000$; within each row $\rho \in \{0.5, 0.7\}$.

As seen in Figure 6, LP(4)'s RMSE matches that of a VAR of order $\approx$ 18–19 in both the low ($\rho = 0.5$) and mid ($\rho = 0.7$) persistence scenarios in small samples. Notably, the parsimonious VAR(1) and VAR(2) are among the lowest-RMSE specifications, as their low variance masks an underlying bias from insufficient lags. There is no visible difference in performance between nonlinearity strengths at low persistence levels, irrespective of sample size, as the misspecification requires sufficient persistence to bite. When moving to medium persistence, RSME rises with γ across all VAR(p) and LP(4) models.

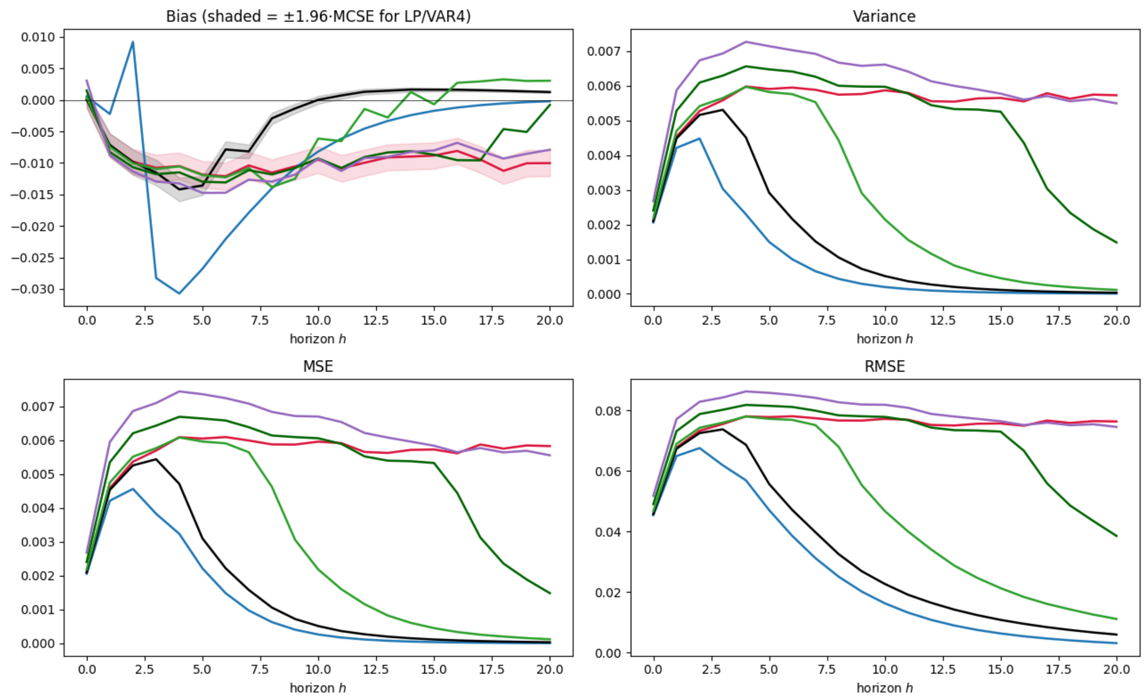
Next, the asymptotic approximation removes the variance that masked the bias of the parsimonious VAR(1) and VAR(2) models. Both reveal the wrong-estimand effect by blowing up and significantly underperform LP(4), while VAR(3)/(4) become the lowest-RMSE specification. The VAR sweep also converges to LP(4) performance in higher lags, irrespective of persistence. Lastly, the RMSE gap between high and low γ shrinks with sample size, consistent with the nonlinearity acting through inflated and asymptotically vanishing variance rather than bias.

Figure 7: Sample Point Estimates under Strong Nonlinearity

(a) $\rho = 0.5$



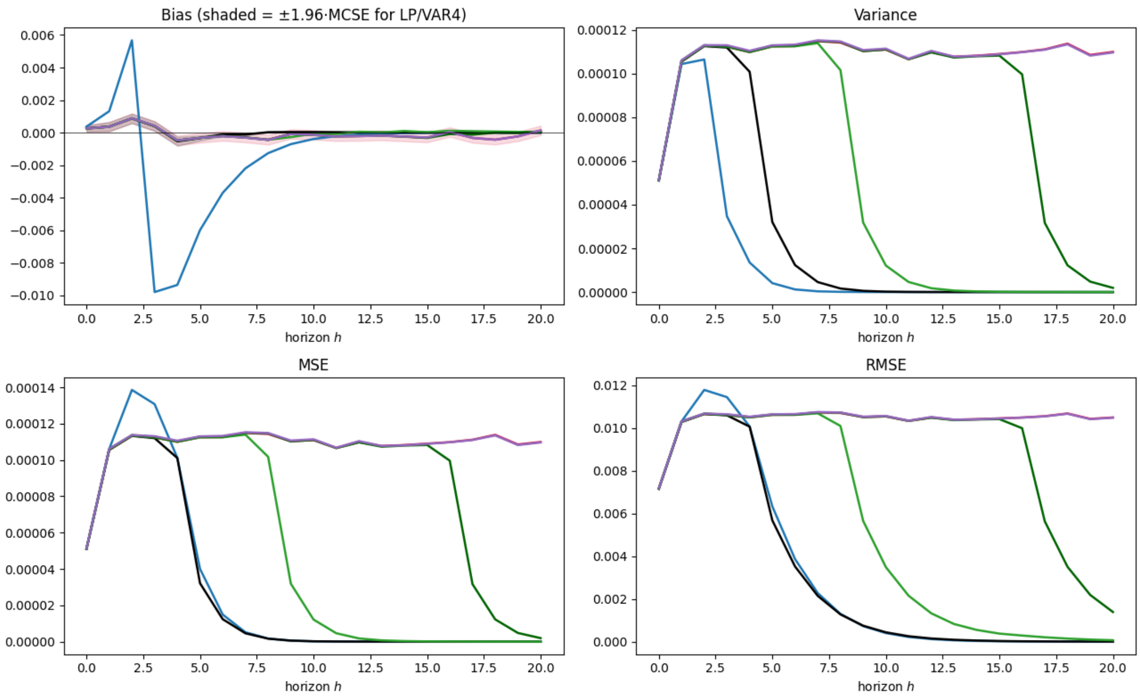
(b) $\rho = 0.7$



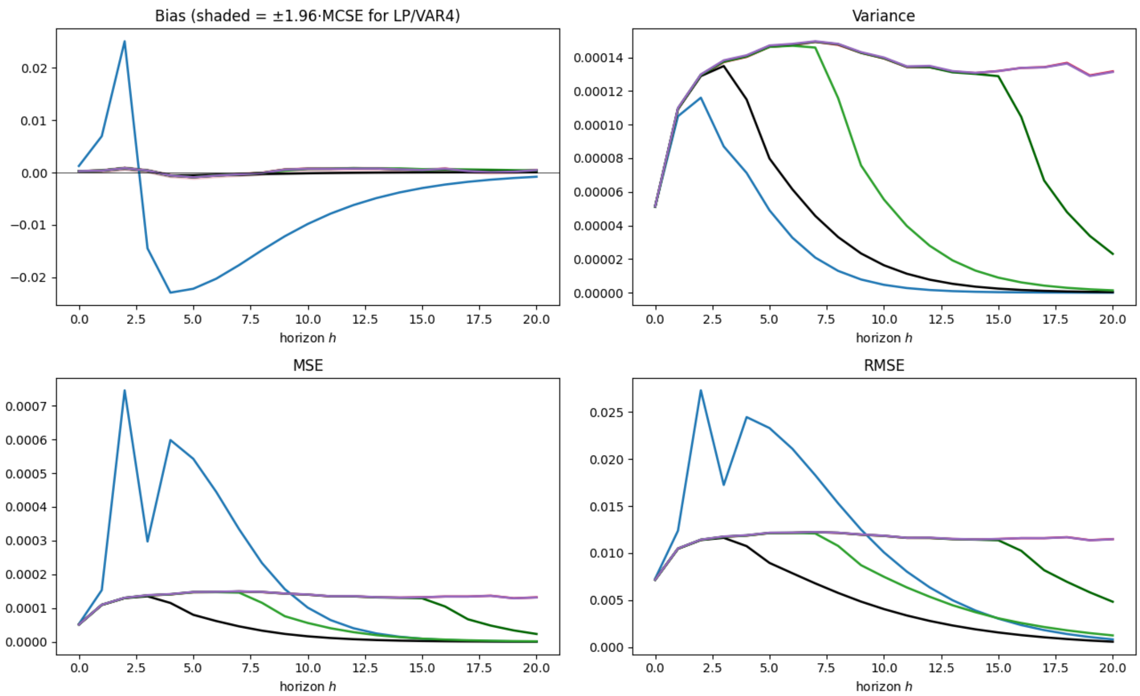
Note. Bias, variance, MSE, and RMSE of $\hat{\theta}_h$ relative to the true (linear-projection) estimand θ_h for the nonlinear lag-quadratic DGP, at $T = 250$, $B = 5{,}000$, $\gamma = 0.6$. LP(4) (red) is compared against a representative VAR(q) set: VAR(2) blue, VAR(4) black, VAR(8) light green, VAR(16) dark green, VAR(23) purple. Rows: $\rho \in \{0.5, 0.7\}$; the shaded band on bias is the 95% Monte Carlo standard-error band.

Figure 8: Asymptotic Point Estimates under Strong Nonlinearity

(a) $\rho = 0.5$



(b) $\rho = 0.7$



Note. As in the sample point-estimation figure but at $T = 10{,}000$ (approximating the asymptotic limit), $B = 5{,}000$, $\gamma = 0.6$. Estimators: LP(4) red, VAR(2) blue, VAR(4) black, VAR(8) light green, VAR(16) dark green, VAR(23) purple. Rows: $\rho \in \{0.5, 0.7\}$; the shaded band on bias is the 95% Monte Carlo standard-error band.

Moving to point prediction (Figures 7 and 8), the equivalence under a nonlinear DGP becomes even clearer. In the small sample, bias in the low and mid persistence scenarios is noisy, manifesting as oscillating lines. VAR bias tracks LP bias up to a horizon within the VAR's lag order before converging to zero. VAR(2) violates this and oscillates strongly, peaking around horizon 3, then converging to zero. LP exhibits a slightly oscillating but roughly constant negative bias.

The key insight appears asymptotically, as every model except VAR(2) collapses onto the zero line, irrespective of persistence. VAR(2) shows large oscillation, peaking at an absolute bias of ≈ 0.01 and ≈ 0.025 (low and mid persistence respectively) before decaying to zero. Its insufficient lag specification targets a different linear projection, which manifests itself in bias when compared to the linear representation utilized by LP(4) and higher order VARs.

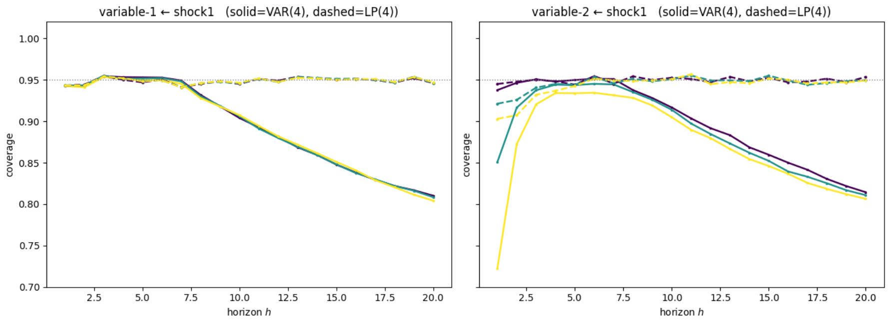
In variance, the lower-order VARs trace LP closely, then peak and fall toward zero roughly one horizon before their lag order q, while VAR(23) ($q > 20$) never turns within the $h \leq 20$ window. Larger-order VARs exceed LP variance at early horizons, whereas LP variance oscillates softly around a constant level. Higher persistence keeps the IRF, and hence the variance, elevated for longer, giving the more rounded, less horizontal profiles. Asymptotically the pattern sharpens, as all models trace LP variance almost exactly, dipping about one horizon before their lag order, then converging to zero.

RMSE and MSE mirror the variance pattern almost exactly, with the exception of VAR(2) which blows up at short horizons, exceeding the other specifications by a wide margin. Since VAR(2) blows up in bias, MSE and RMSE yet shows no anomaly in variance, the evidence again points to a wrong-estimand effect from insufficient lag length, as with MSE = bias^2 + variance, a bias-only blow up cannot surface in the variance panel.

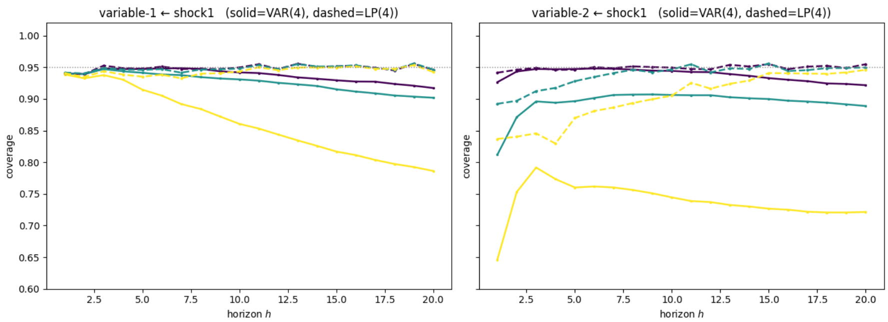
Both the bias and the variance of a VAR(q) turn down at horizons tied to its lag order, because its impulse response is built by iterating the estimated coefficient matrix. While $h \leq q$, the horizon-h response $\Psi_h = \sum_{j=1}^{q} \hat{A}_j \Psi_{h-j}$ still draws directly on the estimated lag matrices, so both its bias and its sampling variance stay elevated. Once $h > q$ the response is pure iteration of the estimated companion, which is stationary, so each further step multiplies the estimate by a contraction and both bias and variance damp geometrically from $h \approx q$ onward. Higher-order VARs therefore turn down later, and VAR(23), with $q > 20$, never drops within the $h \leq 20$ window. LP, by contrast, estimates each horizon by a separate direct regression and never iterates, so its bias and variance stay roughly flat across all horizons. The staggered turning points are thus a consequence of iteration, not of the estimand decaying or of VAR performance improving at higher horizons.

Figure 9: Confidence-Interval Coverage under Nonlinear Lag Misspecification

(a) $\rho = 0.5$



(b) $\rho = 0.7$



Note. Empirical coverage of nominal 95% CIs by horizon ($h = 1, \dots, 20$) for the nonlinear lag-quadratic DGP at $T = 10{,}000$, $B = 5{,}000$. *Left:* response of variable 1 to shock 1; *right:* variable 2 to shock 1. Within each panel VAR(4) is solid (square markers) and LP(4) dashed (circle markers), one colour per nonlinearity strength: purple $\gamma = 0$ (baseline), teal $\gamma = 0.3$, yellow $\gamma = 0.6$; the dotted line marks the 95% reference. VAR(q) intervals use the delta method, LP(4) heteroskedasticity-robust SEs. Rows: $\rho \in \{0.5, 0.7\}$. VAR under-covers as γ grows while LP holds near nominal.

Lastly, the coverage analysis (Figure 9) shows clear evidence of differing robustness to heteroskedasticity between the two estimators. At low persistence, LP(4) coverage sits near the nominal 95% level for both variables, with only slight deviation at low horizons, while VAR(4) coverage develops a mild negative slope after about horizon 7. The γ levels barely differ at later horizons and show only some reduced coverage at low horizons. At mid persistence, LP(4) is almost unaffected for variable 1 (a slightly larger deviation at low horizons), and for variable 2 it dips at low horizons before recovering to near nominal at longer horizons. VAR(4) behaves differently: for low and medium γ it attains *higher* coverage than under low persistence in variable 1, whereas high γ now produces reduced coverage, with the negative slope beginning at earlier horizons across both variables. Variable 2 VAR(4) exhibits a stronger fanning of the γ levels, so the three curves no longer overlay.

As LP defaults to heteroskedasticity-robust (HC) standard errors while the VAR(4) intervals are built from the homoskedastic delta method, we expect VAR(4) to under-cover under the heteroskedasticity induced by the nonlinear DGP. We observe exactly that, with one apparent exception: VAR(4) coverage *improves* with persistence at low γ. This reflects two opposing persistence effects. First, the VAR(4) interval under-covers at long horizons even at $\gamma = 0$, where no heteroskedasticity is present (visible already at $\gamma = 0$, where variable-1 coverage rises from ≈ 0.89 to ≈ 0.94 between the two persistence levels): once the impulse response has decayed toward zero, the delta-method standard error becomes too small relative to the true sampling spread and the interval is too narrow—the negative slope after horizon 7. Because higher persistence keeps the response large for longer, this long-horizon collapse is pushed out of the window, *raising* average coverage. Second, the genuine heteroskedasticity channel *lowers* coverage and *strengthens* with persistence, as a more variable state amplifies the $y_{1,t-1}^2$ variance term. At low and medium γ the first effect dominates and coverage rises with persistence; at $\gamma = 0.6$ the heteroskedasticity channel overwhelms it and coverage falls below the low-persistence level, most severely for variable 2, which loads directly on the heteroskedastic innovation.

4.3 Discussion of Results & Conclusion

The results confirm that the population equivalence of Plagborg-Møller & Wolf (2021) survives nonlinear functional misspecification. Because the centred lag-quadratic is approximately orthogonal to the linear lag space, it leaves the best linear projection essentially unchanged. This results in the estimands being nearly γ-invariant, and both LP(4) and VAR(4) recoverering it asymptotically with no bias floor. The only systematic departures come from the under-lagged VAR(1) and VAR(2), which target a different linear projection and retain a non-vanishing bias. This is not a violation of the equivalence but its precondition, as the result is defined to hold at sufficient lag length.

This split between population and finite-sample behaviour is not a contradiction but a direct consequence of what the equivalence asserts. Plagborg-Møller & Wolf (2021) equate the two estimands, so the probability limits both procedures converge to, rather than the estimates themselves. Both are functions of the same population second moments, and at those true moments the direct projection of LP and the iterated coefficients of the VAR are merely two algebraic routes to the same impulse response. In a finite sample, however, those moments are replaced by noisy estimates, and the two routes propagate that noise differently. VARs compound it through the iterated companion, while LP estimates each horizon by a separate regression. Thus, since the impulse response is a nonlinear function of the coefficients, each route also carries its own finite-sample bias. The estimators therefore share a target but not a sampling distribution, and they coincide only in the limit. Thus, the two estimators divergence along two dimensions when appliede to sample data, namely point estimation and inference.

In point estimation, VAR(4) retains the MSE advantage. Since both estimators carry the same small bias, their difference appears in their variance. Small sample volatility leads to LP's direct per-horizon regressions to include more noise than the VAR's iterated estimates.

In inference, the ranking reverses, as the lagged-state nonlinearity injects conditional heteroskedasticity into the projection residuals, which the homoskedastic delta-method interval of the VAR cannot accommodate. VAR(4) thus under-covers increasingly with γ and persistence, while LP(4)'s heteroskedasticity- robust interval holds near the nominal level.

These coverage differences concern inference rather than the point estimand, and therefore do not contradict the equivalence. LP(4) and VAR(4) still target the same impulse response, and the same linear representation of the non-linear DGP. The broader lesson is that estimand equivalence does not imply inferential equivalence, as two estimators of the same object can posess differing interval validity once their standard-error assumptions are violated.

Thus, in a misspecification based on a non-linear DGP, LP's robustness yields a genuine advantage, but it is an advantage in inference under default standard errors, not in point estimation. When the misspecification leaves the required lag order unchanged, the choice between the two estimators is therefore objective-dependent. VAR for the efficiency of the point estimate, LP for valid inference under heteroskedasticity. Lastly, VAR models can also be augmented to leverage robust standard errors, which would reduce the found supremacy of LPs regarding SE coverage.

In summary, Extension 2 shows that the population equivalence of LP and VAR is robust to functional misspecification through a non-linear DGP, provided the fitted lag order remains sufficient. The nonlinearity leaves the shared estimand almost untouched and is felt only in the second moment of the residuals, which produces a clean division of labour between the two estimators. VARs archieve lower variance point estimation, while the LP delivers the only valid inference once heteroskedasticity is present. This is the first design in which Local Projections' robustness translates into a tangible advantage. Still, this advantage is only confined, as it is restricted to inference and can be partly recovered by equipping the VARs with robust standard errors.

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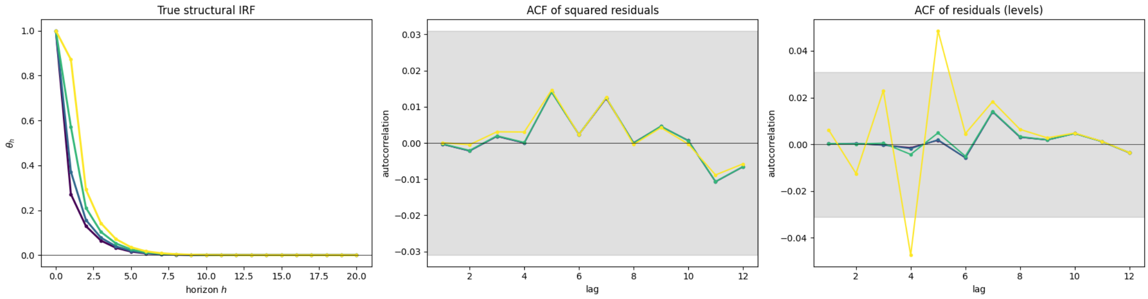
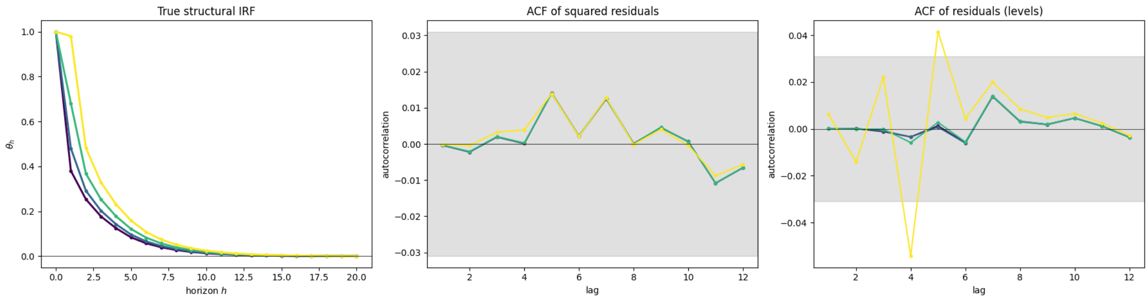
Appendix

Figure A.1: True IRF Functions by MA Misspecification Strength

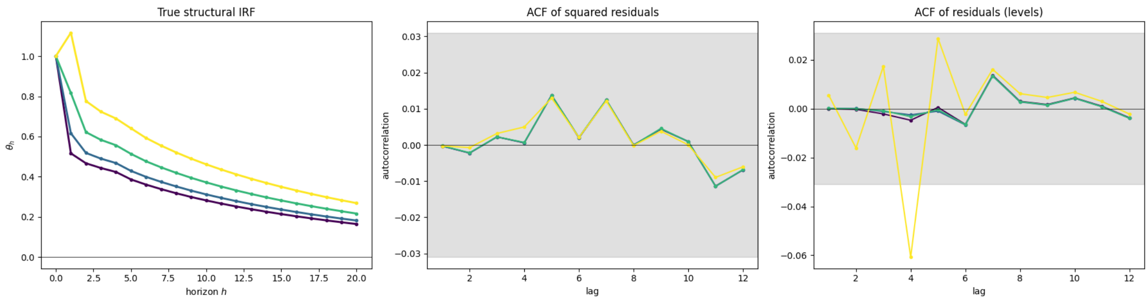
(a) $\rho = 0.5$



(b) $\rho = 0.7$



(c) $\rho = 0.95$



Note. Each subfigure ($\rho \in \{0.5, 0.7, 0.95\}$) has three panels: *(left)* the true structural IRF θ_h; *(middle, right)* the autocorrelation of the squared and level residuals of a VAR(4) fit to the VARMA data. The squared-residual ACF stays inside the band (conditionally homoskedastic), while the level-residual ACF breaks out as θ grows — the MA serial correlation the equal-lag VAR(4) cannot absorb. Lines: purple $\theta = 0$ (baseline), blue $\theta = 0.1$, green $\theta = 0.3$, yellow $\theta = 0.6$; the grey area is the 95% white-noise band.

Declaration of AI Usage

During the presentation of this seminar paper, I Patrick Tenner/Aaron Liebig, used Claude for drafting and coding assistance, as well as Gemini for linguistic smoothing. I have reviewed and edited the content as needed and take full responsbility.

Patrick Tenner/Aaron Liebig